

REPEATED SUMMANDS IN EULER’S SIXTH-POWER EQUATION

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ABSTRACT. We study the sixth-power equation

$$a_1^6 + a_2^6 + a_3^6 + a_4^6 + a_5^6 = B^6$$

in positive integers without assuming that the five left-hand bases are distinct. In every primitive solution, we prove that B is coprime to 42, that exactly one left-hand occurrence is a unit at each of 2, 3, and 7, and consequently that every repeated base is divisible by 42. We then prove that at least three distinct left-hand bases are necessary. The only noncongruential case is the multiplicity pattern (4, 1); a coprime half-sum/half-difference factorization reduces it to Fermat’s Last Theorem for exponent three. We also classify the forced divisibility in the two exactly-three-distinct patterns and certify by exact enumeration that no repeated-summand solution has $B \leq 1000$. Finally, a source-level audit shows that a current public GPU campaign imposes pairwise distinctness, so its reported bounds do not by themselves cover the repeated-summand strata. The unrestricted five-summand equation remains open.

1. INTRODUCTION

Euler’s sum-of-powers conjecture asserts that a k th power cannot be expressed as a sum of fewer than k positive k th powers. Lander and Parkin disproved the assertion for $k = 5$ with the exact identity

$$27^5 + 84^5 + 110^5 + 133^5 = 144^5$$

[3]; Elkies later disproved the quartic case [1]. The equal-sums notation and the early computational frontier are surveyed in [4].

For exponents larger than five no counterexample is currently known. The FrontierMath problem asks for nonzero integers and does not impose distinctness [2]. For exponent six, signs disappear after taking absolute values, and its tight equation is

$$(1) \quad a_1^6 + a_2^6 + a_3^6 + a_4^6 + a_5^6 = B^6, \quad a_i, B \in \mathbb{Z}_{>0}.$$

Repeated bases are therefore part of the stated problem.

This point matters when interpreting computational records. Resta and Meyrignac report on page 1054 that no solution with right-hand base below

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730,000 had been found for the equation with *six* sixth-power summands [5]. That peer-reviewed sentence is not literally a bound for the five-summand equation (1). Separately, a current public GPU repository reports substantial searches for (1), but its mathematical domain and final exact guards require the five bases to be pairwise distinct [6]. We treat these as scope facts, not as criticism of a deliberately narrower computation.

Our main result addresses the omitted strata.

Theorem 1.1. *Let $(a_1, \dots, a_5, B)$ be a primitive positive integer solution of (1), with repetitions allowed. Then:*

- (1) $\gcd(B, 42) = 1$;
- (2) *exactly one occurrence among the a_i is odd, exactly one is not divisible by 3, and exactly one is not divisible by 7*;
- (3) *every base value occurring at least twice among the a_i is divisible by 42*;
- (4) *at least three distinct values occur among $a_1, \dots, a_5$.*

We did not locate this explicit repeated-summand statement in the sources and exact-equation searches recorded with the paper. This is a bounded novelty report, not a claim of priority. The theorem is a partial result: it neither constructs a counterexample nor proves that (1) has no solution.

2. PRIMITIVE NORMALIZATION AND UNIT OCCURRENCES

Because (1) is homogeneous, any solution can be divided by the common gcd of its six bases. Thus no generality is lost by assuming primitivity. Positivity also gives $a_i < B$ for every i .

Lemma 2.1 (Unique unit occurrences). *In a primitive solution of (1), B is coprime to 42. Among the five left-hand occurrences, exactly one is odd, exactly one is coprime to 3, and exactly one is coprime to 7.*

Proof. For every integer x ,

$$x^6 \equiv \begin{cases} 0, & 2 \mid x, \\ 1, & 2 \nmid x, \end{cases} \pmod{8}, \quad x^6 \equiv \begin{cases} 0, & 3 \mid x, \\ 1, & 3 \nmid x, \end{cases} \pmod{9},$$

and

$$x^6 \equiv \begin{cases} 0, & 7 \mid x, \\ 1, & 7 \nmid x \end{cases} \pmod{7}.$$

Use in turn $(p, m) = (2, 8), (3, 9), (7, 7)$, and let t_p be the number of left-hand occurrences not divisible by p . If $p \mid B$, then $t_p \equiv 0 \pmod{m}$. Since $0 \leq t_p \leq 5 < m$, this forces $t_p = 0$; all six bases are then divisible by p , contradicting primitivity. Hence $p \nmid B$, so $t_p \equiv 1 \pmod{m}$, and the same range gives $t_p = 1$. $\square$

Corollary 2.2 (Repeated-base divisibility). *Every left-hand base having multiplicity at least two is divisible by 42.*

Proof. If a repeated value were a unit at one of 2, 3, or 7, its repeated occurrences would give at least two unit positions for that prime, contrary to Lemma 2.1. It is therefore divisible by all three primes. $\square$

3. EXCLUDING ONE- AND TWO-BASE MULTISSETS

The partitions of five into at most two multiplicities are

$$(5), \quad (4, 1), \quad (3, 2).$$

Corollary 2.2 immediately excludes (5) and (3, 2): in either case every occurrence belongs to a repeated block, so every a_i is divisible by 42, contradicting Lemma 2.1. The remaining pattern uses one additional global ingredient.

Proposition 3.1. *The equation*

$$(2) \quad 4a^6 + b^6 = B^6$$

has no primitive solution in positive integers.

Proof. Modulo 2, the integers b and B have the same parity. If both were even, then $64 \mid B^6 - b^6 = 4a^6$, hence a would also be even, contrary to primitivity. Thus b and B are odd. Moreover, any odd prime dividing both b and B would, by (2), divide a ; hence $\gcd(b, B) = 1$.

Set

$$r = \frac{B^3 - b^3}{2}, \quad s = \frac{B^3 + b^3}{2}.$$

These are positive integers and

$$rs = \frac{B^6 - b^6}{4} = a^6.$$

A common divisor of r and s divides both $r + s = B^3$ and $s - r = b^3$. Consequently $\gcd(r, s) = 1$. Unique factorization in $\mathbb{Z}$ now gives $r = u^6$ and $s = v^6$ for positive integers u, v . Therefore

$$B^3 = r + s = u^6 + v^6 = (u^2)^3 + (v^2)^3,$$

which contradicts Fermat's Last Theorem for exponent three [8, 7]. $\square$

The three multiplicity partitions above are now excluded, proving the last part of Theorem 1.1; its first three parts are Lemma 2.1 and Corollary 2.2.

4. THE EXACTLY-THREE-DISTINCT BOUNDARY

The next unresolved stratum has a complete elementary divisibility classification.

Corollary 4.1. *Suppose a primitive solution of (1) has exactly three distinct left-hand bases.*

- (1) *In pattern (3, 1, 1), the tripled base is divisible by 42. For each $p \in \{2, 3, 7\}$, exactly one of the two singleton bases is divisible by p and the other is not.*

- (2) In pattern $(2, 2, 1)$, both doubled bases are divisible by 42, while the singleton and B are coprime to 42.

Proof. These are the only partitions of five into three positive parts. Every repeated block is divisible by 42 by Corollary 2.2. The remaining claims follow by placing the unique unit occurrence for each of 2, 3, and 7 among the singleton blocks, using Lemma 2.1. $\square$

For reference, Table 1 separates the solved and open multiplicity strata.

TABLE 1. Multiplicity patterns among the five left-hand bases.

Distinct bases	Patterns	Status here
1	(5)	impossible modulo 7, 8, 9
2	(4, 1), (3, 2)	impossible; Proposition 3.1 needed for (4, 1)
3	(3, 1, 1), (2, 2, 1)	open, with Corollary 4.1 restrictions
4	(2, 1, 1, 1)	open; doubled base is a multiple of 42
5	(1, 1, 1, 1, 1)	open distinct-summand stratum

5. EXACT FINITE SEARCH AND SCOPE AUDIT

5.1. Repeated-summand search through $B = 1000$.

Proposition 5.1. *There is no positive integer solution of (1) with a repeated left-hand base and $B \leq 1000$.*

Certificate description. Any solution would scale down to a primitive one. Select two occurrences of a repeated base a . Corollary 2.2 gives $a = 42k$, and the equation becomes

$$2a^6 + b^6 + c^6 + d^6 = B^6, \quad \gcd(B, 42) = 1.$$

The verifier indexes every unordered pair (b, c) with $1 \leq b \leq c < 1000$ by the exact integer $b^6 + c^6$. For every admissible B , every multiple a of 42 below B , and every $d < B$, it tests the exact target

$$B^6 - 2a^6 - d^6$$

against this index, retaining the minimum possible maximum pair entry so that $b, c < B$ is checked without approximation. It indexes 499,500 distinct pair sums and tests 1,879,634 targets. None matches. The one-command verification is

`python code/search_repeated_up_to_1000.py.`

$\square$

This finite proposition is a regression and coverage certificate, not a new height record.

5.2. Audit of a public distinct-summand campaign. At repository commit 88111f2d599da494a68500326d5f1511172a2894, the public problem note states $1 \leq a_1 < \dots < a_5 < B$. Both the CPU and CUDA release solvers sort a candidate tuple and reject it when two adjacent entries are equal [6]. The audit script verifies the commit and these guards directly. Thus the campaign’s reported bounds concern the distinct-summand stratum in the last row of Table 1; they do not by themselves certify the three repeated-summand rows.

This distinction also suggests a cheaper next search. A repeated candidate has a designated 42-multiple, so a meet-in-the-middle implementation should search

$$2(42k)^6 + b^6 + c^6 + d^6 = B^6$$

separately, rather than embedding repetitions in a five-distinct tuple engine. Corollary 4.1 gives still smaller specialized searches for the (3, 1, 1) and (2, 2, 1) strata.

6. REPRODUCIBILITY, LIMITATIONS, AND DISCLOSURE

The machine-readable theorem data are in `certificates/repeated_summands.json`. The primary residue verifier, an independent divisibility-mask enumeration, the factorization audit, the finite search, and the public-code scope audit are run together by

`python code/run_all_verifiers.py.`

All decisive arithmetic uses Python integers. The factorization program is a diagnostic check of the algebra; the unbounded exclusion of (4, 1) is the human proof in Proposition 3.1, with Fermat’s theorem as its explicit external dependency. No proof assistant was used.

The terminal status is *partial theorem*. No six-integer certificate for (1) was found, no unrestricted nonexistence proof is claimed, and the open rows of Table 1 remain genuine. In particular, the paper does not turn either the $B \leq 1000$ computation or a distinct-only external campaign into an assertion about all positive integers.

AI assistance was used for literature organization, conjecture generation, exact-program drafting, adversarial checks, and manuscript preparation. The resulting claims were rederived from the definitions, checked by independent exact programs where applicable, and bounded to the theorem stated above. The authors remain responsible for the mathematical content and citations.

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